Algebra 1 Name Answer Key ____________
Unit 10 Date____________________
Lesson H1 Homework Period___________________

For questions 1-2, consider the absolute value inequality $|4x - 9| - 8 < 19$.

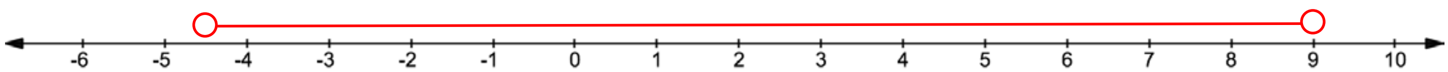
1. Solve the inequality.

$|4x - 9| - 8 < 19$
$\quad +8 \quad +8$
$|4x - 9| < 27$
$-4.5 < x < 9$

$4x - 9 < 27$
$\quad +9 \quad +9$
$4x < 36$
$x < 9$

$4x - 9 > -27$
$\quad +9 \quad +9$
$4x > -18$
$x > -4.5$

2. Graph the solution on the number line.



For questions 3-4, consider the absolute value inequality $2|x - 5| - 4 < 0$.

3. Solve the inequality.

$\frac{2|x-5|}{2} < \frac{4}{2}$

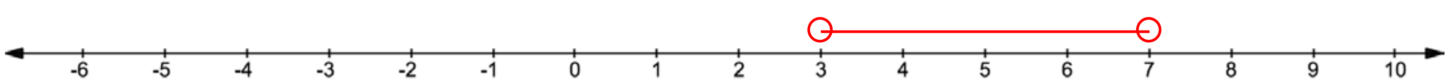
$|x - 5| < 2$

$x - 5 < 2$
$\quad +5 \quad +5$
$x < 7$

$x - 5 > -2$
$\quad +5 \quad +5$
$x > 3$

$3 < x < 7$

4. Graph the solution on the number line.



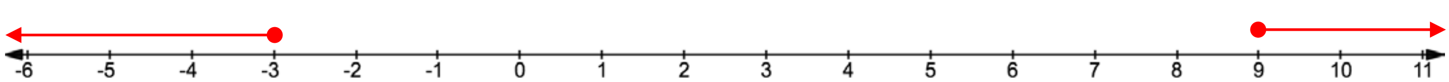
For questions 5-6, consider the absolute value inequality $|3x - 9| \geq 18$.

5. Solve the inequality.

$3x - 9 \geq 18$
$\quad +9 \quad +9$
$3x \geq 27$
$x \geq 9$

$3x - 9 \leq -18$
$\quad +9 \quad +9$
$3x \leq -9$
$x \leq -3$

6. Graph the solution on the number line.



For questions 7-10, solve the absolute value inequality.

7. $-8|x-7| > 32$

$\overline{-8} \quad \overline{-8}$

$|x-7| < -4$

No solution

8. $-3|x-5| - 6 \leq 9$

$+6 \quad +6$

$\frac{-3|x-5|}{-3} \leq \frac{15}{-3}$

$|x-5| \geq -5$

Infinitely many solutions

9. $|x-11| \geq -3$

Infinitely many solutions

10. $|7x+4| - 6 < 8$

$+6 \quad +6$

$|7x+4| < 14$

$7x + 4 < 14$

$-4 \quad -4$

$\frac{7x}{7} < \frac{10}{7}$

$x < 1\frac{3}{7}$

$7x + 4 > -14$

$-4 \quad -4$

$\frac{7x}{7} > \frac{-18}{7}$

$x > -2\frac{4}{7}$